Introduction

Purpose

This document specifies the requirements for CalvinFinder, a mobile app that helps users find classes, buildings and points of interest (POIs) around Calvin University. It defines what the system must be able to do and what qualities it must have so the team shares one vision for what the product should like before, during and after the development process.

Intended Audience

Developer team: Team G
Course Instructor: Prof. Victor Norman
Stakeholders: Team H
Future maintenance team

Intended Use

CalvinFinder is intended for anyone visiting or navigating Calvin University who needs to locate a building, room, or amenity. Typical uses would include an incoming student finding a classroom on the first day of classes, a Calvin guest trying to locate a building or anyone looking to find the nearest printer, bathrooms or vending machine. It is used to reference and help an individual find the way.

Product Scope

CalvinFinder is a mobile app that shows an interactive campus map, lets users select buildings to view their layouts, floors and displays POIs inside the buildings. Unlike other campus navigation tools, it combines classroom locations and POIs on one platform.

Definitions and Acronyms

Term	Definition
POI	Point of Interest such as vending machine, printer,water fountain, bathroom
Building Layout	A floor plan showing rooms and features in the building
Room	A number space in the building (e.g., SB 010 in Science Building)
Calvin ID	A Calvin University user account number
Guest	Someone using the app without signing in
Admin	A person who maintains the data
SRS	Software Requirements Specification
UI	User Interface

MVP	Minimum Viable Product
FR/NFR	Functional / Non-functional requirement

Overall Description

Calvin Finder is a campus navigation app for Calvin University.

User Needs
Primary users: incoming students. These are newly admitted or first-year students who are enrolled members of the university but have little familiarity with the campus. They use the app to find classrooms, residence halls, dining locations, and offices, and to learn the layout over their first weeks. Getting lost makes them late, adds stress to an already stressful transition, and discourages them from asking for help.

Secondary users: visitors. These are external users with no formal role in the university, such as visiting parents and prospective students touring campus. They spend a short time on campus and use the app to reach specific destinations such as admissions, parking, and event venues without needing a guide. Getting lost costs them time and gives a poor first impression of the university.

Dependencies. Calvin Finder is a mobile app and depends on the following to function:

- **Mobile framework:** React Native, and the iOS and Android versions it supports.
- **Backend:** Supabase for the database and data storage, which requires an active Supabase project and network access to it.
- **Campus data:** Floor plans provided by Calvin University in PDF format. The team will manually convert these into nodes, connections, and room labels for routing, so the app depends on the plans being accurate and on the team completing this conversion.
- **Third-party libraries:** React Native packages for map rendering, navigation and routing, and UI, and their continued compatibility with the framework version used.
- **Internet connectivity:** The app requires an internet connection (Wi-Fi or cellular data) to retrieve campus data and does not support offline use.
- **Reused components:** No software is reused from previous projects.

Assumptions.

- The app does not use GPS or real-time location tracking. Users select their starting location manually, and routes are computed from that point.
- Users have a smartphone with an active internet connection while using the app.
- Calvin's floor plans are accurate at release, and campus changes are infrequent enough to be handled by manual updates.
- Supabase remains available and its usage limits are sufficient for the project's expected load.

Systems Functions and Requirements

Functional Requirements

FR-1 The user shall be able to view the campus map so that they can understand the campus layout.

FR-2 The user shall be able to pan and zoom the map so that they can look at specific areas in detail.

FR-3 The user shall be able to select a building on the map so that they can see its details.

FR-4 The user shall be able to see building names when the hover over them so that they can identify buildings without selecting them.

FR-5 The user shall be able to recenter the map with one tap so that they do not get lost after panning.

FR-6 The user shall be able to see their current location on the map so that they know where they are relative to their destination.

FR-7 The user shall be able to switch between illustrated and standard map views so that they can choose the view they prefer.

FR-8 The user shall be able to select a building and view its floor layouts so that they can find locations inside the building.

FR-9 The user shall be able to switch between floors so that they can view different levels of a multi-story building.

FR-10 The user shall be able to view room numbers on floor layouts so that they can identify specific rooms.

FR-11 The user shall be able to view entrances, stairs, elevators, and restrooms on floor layouts so that they can orient themselves within a building.

FR-12 The user shall be able to view a building's operating hours so that they know whether the building is open.

FR-13 The user shall be able to view which building entrances are accessible so that they can plan an accessible route.

FR-14 The user shall be able to view points of interest on the campus map so that they can find campus amenities.

FR-15 The user shall be able to select a point of interest so that they can view its name, type, and location details.

FR-16 The user shall be able to filter points of interest by category so that they only see locations relevant to their needs.

FR-17 The user shall be able to view points of interest within buildings on floor layouts so that they can find them indoors.

FR-18 The user shall be able to view the nearest point of interest of a selected type so that they can find it quickly.

FR-19 The user shall be able to report a missing or incorrect point of interest so that the data remains accurate.

FR-20 The user shall be able to search for buildings by name so that they can find them quickly.

FR-21 The user shall be able to search for rooms by room number so that they can locate specific rooms.

FR-22 The user shall be able to search for points of interest by category so that they can find desired amenities.

FR-23 The user shall be able to view search suggestions while typing so that they do not need exact spelling.

FR-24 The user shall be able to view recent searches so that they can repeat common lookups.

FR-25 The user shall receive suggested alternatives when no search results are found so that they can refine their search.

FR-26 The user shall be able to find locations using abbreviations or minor misspellings so that small mistakes do not prevent successful searches.

FR-27 The user shall be able to browse a list of buildings so that they can explore campus locations.

FR-28 The user shall be able to expand a building entry so that they can view its rooms.

FR-29 The user shall be able to select a location from the directory so that they can jump to that location.

FR-30 The user shall be able to drag the location panel so that they can balance map and directory visibility.

FR-31 Calvin students and staff shall be able to sign in using their Calvin ID so that they can access personalized features.

FR-32 Visitors shall be able to continue as guests so that they can use the application without a Calvin account.

FR-33 The user shall be able to reset a forgotten password so that they can regain access to their accounts.

FR-34 The user shall receive a clear error message when credentials are incorrect so that they know what needs to be corrected.

FR-35 The user shall be able to remain signed in between sessions so that they do not need to log in repeatedly.

FR-36 The user shall be able to sign out so that their account remains secure on shared devices.

FR-37 The user shall be able to save a location as a favorite so that it can be accessed quickly later.

FR-38 The user shall be able to view favorite locations so that frequently used places are easy to access.

FR-39 The user shall be able to remove a favorite so that the favorites list remains relevant.

FR-40 Signed-in users shall be able to synchronize favorites across devices so that favorite locations remain available everywhere they sign in.

FR-41 The user shall be able to open application settings so that they can customize the application.

FR-42 The user shall be able to choose between light mode and dark mode so that the application is comfortable to view.

FR-43 The user shall be able to adjust text size and units so that the application is easy to read.

FR-44 The user shall be able to manage location and notification permissions so that they can control their privacy preferences.

FR-45 The user shall be able to view the application's privacy policy so that they understand how their data is used.

FR-46 The user shall be able to submit feedback about application errors, incorrect information, or usability issues so that administrators can improve the application.

FR-47 Administrators shall be able to update campus map information, room information, building information, and points of interest so that application data remains accurate.

Non-Functional requirements

NFR-1 The campus map shall load within 3 seconds on a typical mobile connection.

NFR-2 Search results shall appear within 1 second of a user query.

NFR-3 Floor changes and POI filtering shall update within 1 second.

NFR-4 Only authenticated users shall be able to access administrator functions.

NFR-5 Passwords shall never be stored in plain text.

NFR-6 Data transmitted between the application and server shall be encrypted.

NFR-7 User sessions shall expire after a period of inactivity.

NFR-8 A first-time user shall be able to locate a building without requiring instructions.

NFR-9 The app shall display a loading indicator while data is being loaded.

NFR-10 The app shall inform users when location information is incomplete or unavailable.

NFR-11 The app shall allow access to previously loaded map and building data while offline.

NFR-12 The app shall display a clear message when network connectivity is unavailable or data cannot be loaded.

NFR-13 The app shall allow users to retry failed data requests without restarting the application.

NFR-14 Approved map updates shall be made available to users within 24 hours of approval.

NFR-15 Buildings, rooms, and points of interest shall provide meaningful labels for screen readers.

NFR-16 The application shall support adjustable text sizes and high-contrast display options.

NFR-17 Information shall not rely solely on color for interpretation.

NFR-18 Interactive touch targets shall be large enough to support users with limited dexterity.

NFR-19 The application shall support filtering for accessible entrances, elevators, and ramps.

System Features
Campus Map, Building Layouts, POIs, Search and Browse, Accounts and Favorites, Feedback & Admin, Error handling

Other Requirements

- **Database will hold:**
    - Map layouts
    - Names of rooms
    - POIs
    - User schedules
    - Routes for pathing

Relationships: A building has many floors, a Floor has many rooms, A POI belongs to one category and may belong to a Building or Floor. A user has many favorites and may submit many issue reports.